Ada Mid-term assignment

Background:

You will be implementing a generic "Database" and a generic "Controller". You have been given a main.adb test file. The Controller serves as the only communication link between the test file and the database (i.e do not import the database into the test file!). The database will be implemented using an array. There will be 2 other packages, a User package and a Message package. You should have 1 implementation of a Controller and Database which can work for both the User and Message objects.

Use the same variable names, function names and package names where specified in red or the test cases will not work!

User Package:

Create a new package for the User which contains a **User** type which is private.
The User has the following attributes:

- **Email**: Unbounded_String
- **Name**: Unbounded_String
- **Surname**: Unbounded_String
- **Age**: Natural

Create a "Constructor" procedure to create a User called **Create_User** which takes in each of the attributes and assigns them to a User. (for the string attributes, the parameter types must be String, not Unbounded_String).

Create Getters for each of the attributes (for the string attributes, the return type must be String, not Unbounded_String).

Create a function which returns a String (not Unbounded_String) which constructs a JSON string from the User object.

Message Package:

Create a new package for the Message which contains a **Message** type which is private.
The Message has the following attributes:

- **ID**: Integer
- **Text**: Unbounded_String
- **Timestamp**: Unbounded_String
- **User_Sent**: User
- **User_Received**: User

Create a "Constructor" procedure to create a Message called **Create_Message** which takes in each of the attributes and assigns them to a Message. (for the string attributes, the parameter types must be String, not Unbounded_String).

Create Getters for each of the attributes (for the string attributes, the return type must be String, not Unbounded_String).

Create a function which returns a String (not Unbounded_String) which constructs a JSON string from the Message object.

Database Package

Create a generic package which has the following generic attributes:

- **Elem** – any nonlimited definite type
- **Key** – any nonlimited type
- A function called **Condition** which takes in a Key and an Elem and returns a Boolean

Create a Database type which is limited private and takes in a parameter called **Max** of type Natural. The Database type contains an array of Elements using a Natural index from **0 to the Max**. The implementation of this array must be private. The Database type also contains a variable which keeps track of the current number of elements in the database.

Create the following procedures:

- **Insert_Elem** -> adds an element to the Database
- **Get_Elem** -> Receives a Key as a parameter and searches for the Element using the Key. It then assigns the Element to an output parameter (This is a procedure – not a function).
- **Update_Elem** -> Takes in a Key and an Element Object. It searches for the Element in the Database using the key and updates it with the new Element.
- **Delete_Elem** -> Takes in a Key and searches for the Element in the Database and removes it.

Use the generic **Condition** function to help with searching.

All of the above procedures have an output parameter called **Success** which is an output parameter. This parameter is of Boolean Type and is set to True when the operation succeeds. (False when the database is full and cannot insert another element, or when searching for an element and the element cannot be found).

Controller Package

Create a generic package which has all the same generic attributes as the Database Package as well as another function called **JSON** which takes in an Element and returns a String

Instantiate the Database Package with the necessary generic conditions.

Create a **Controller** type which is limited private and has a parameter called Max of type Natural. The Controller has a **DB** Variable of type Database from the Database package you have instantiated. Pass the **Max** of Controller to the Database.

Create the following procedures:

- **Create** -> Takes in an Elem and calls the Insert_Elem on the database. Prints a message in case of success and one for failure.
- **Read** -> Takes in a Key and calls the Get_Elem on the database. Prints the JSON of the Elem in case of success and an error message in case of failure.
- **Update** -> Takes in a Key and an Elem and calls the Update_Elem on the database. Prints a message in case of success and one for failure.
- **Delete** -> Takes in a Key and calls the Delete_Elem on the database. Prints a message in case of success and one for failure.

Main (Test File Given)

Import the necessary packages.

Implement the 2 versions of the generic Condition function for the User and Message objects. The User is compared by email, whereas the message is compared by the key. Where Key will = email/ID and Elem = User/Message.

Instantiate a Controller Package for User and one for Message. For each package use the relevant type and function parameters for Elem, Key, Condition and JSON.

Create a User Controller variable called **U_Cntrl** which has a max of 10.
Create a Message Controller variable called **M_Cntrl** which has a max of 10.

Do not under any circumstances import the Database Package into the Main and do not import the User or Message Package into the Database or Controller Package!

Do not modify any of the test cases!

Your output should look similar as the following (Depending on your JSON strings):

```
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
```

Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted
Element successfully inserted

Element could not be inserted - database full

{"email": user1@email.com,"name": name 1,"surname": surname 1,"age": 11},
{"email": user2@email.com,"name": name 2,"surname": surname 2,"age": 12},
{"email": user3@email.com,"name": name 3,"surname": surname 3,"age": 13},
{"email": user4@email.com,"name": name 4,"surname": surname 4,"age": 14},
{"email": user5@email.com,"name": name 5,"surname": surname 5,"age": 15},
{"email": user6@email.com,"name": name 6,"surname": surname 6,"age": 16},
{"email": user7@email.com,"name": name 7,"surname": surname 7,"age": 17},
{"email": user8@email.com,"name": name 8,"surname": surname 8,"age": 18},
{"email": user9@email.com,"name": name 9,"surname": surname 9,"age": 19},
{"email": user10@email.com,"name": name 10,"surname": surname 10,"age": 20},

{"id": 1,"text": some text 1,"timestamp": 2025-03-04 16:05:49,"user_sent": {"email": user1@email.com,"name": name 1,"surname": surname 1,"age": 11},,"user_received": {"email": user7@email.com,"name": name 7,"surname": surname 7,"age": 17},},
{"id": 2,"text": some text 2,"timestamp": 2025-03-04 16:05:49,"user_sent": {"email": user2@email.com,"name": name 2,"surname": surname 2,"age": 12},,"user_received": {"email": user8@email.com,"name": name 8,"surname": surname 8,"age": 18},},
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Element could not be found

Element successfully updated

Element successfully updated

{"email": user1@email.com,"name": name 1,"surname": surname 1,"age": 11},
{"email": user2@email.com,"name": name 2,"surname": surname 2,"age": 12},
{"email": user3@email.com,"name": name 3,"surname": surname 3,"age": 13},
{"email": user4@email.com,"name": name 4,"surname": surname 4,"age": 14},
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Element could not be found

Element could not be found

Element succesfully deleted

Element succesfully deleted

{"email": user1@email.com,"name": name 1,"surname": surname 1,"age": 11},
{"email": user2@email.com,"name": name 2,"surname": surname 2,"age": 12},
{"email": user3@email.com,"name": name 3,"surname": surname 3,"age": 13},
{"email": user4@email.com,"name": name 4,"surname": surname 4,"age": 14},
{"email": user5@email.com,"name": name 5,"surname": surname 5,"age": 15},
{"email": user6@email.com,"name": name 6,"surname": surname 6,"age": 16},
{"email": user7@email.com,"name": name 7,"surname": surname 7,"age": 17},
{"email": user8@email.com,"name": name 8,"surname": surname 8,"age": 18},
{"email": user9@email.com,"name": name 9,"surname": surname 9,"age": 19},
Element could not be found

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Element could not be found

Element could not be found

Element could not be found